

DS-A1 Bike Sharing Data Audit

Data Provenance and Measurement Audit

1. Question

How does hourly bike rental volume vary with weather conditions, working-day status, and hour of day in the 2011-2012 Capital Bikeshare data?

This report uses basic data checks and descriptive statistics. No prediction model is trained.

2. Dataset

The official UCI Bike Sharing hourly file contains 17,379 rows and 17 columns. It covers Capital Bikeshare in Washington, D.C., from 2011-01-01 to 2012-12-31. The raw CSV and official README are included unchanged.

Source: UCI Machine Learning Repository, Hadi Fanaee-T (2013). DOI: 10.24432/C5W894.
License: CC BY 4.0. Download date: 2026-09-30.

The SHA-256 identifies the exact hourly file:

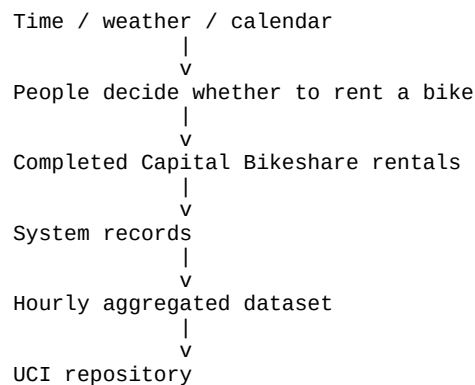
```
e03de4ee4ef4dc376ac6e04bf829673c  
6269e8eba5c60fa121640fa2f829504f
```

3. Research definitions

Term	Definition
Stakeholder	Capital Bikeshare staff who want to understand rental patterns.
Population	Capital Bikeshare system-hours during 2011-2012.
Sample	The 17,379 recorded hours in the hourly CSV.
Unit	One recorded hour for the whole system, identified by date and hour.
Target	cnt: completed rentals in that hour.
Estimand	Mean hourly rental count for different weather conditions, working-day status, and hours of the day in this dataset. $E[\text{cnt} \mid \text{weather}, \text{workingday}, \text{hour}]$.

4. Data-generating process and provenance

The official README describes Capital Bikeshare system records aggregated by hour, with weather information from Freemeteo. The creator prepared these files, which UCI distributes.



Observed rentals may differ from true demand because a person who wants a bike may not always be able to rent one.

Lineage: Capital Bikeshare records + Freemeteo weather -> creator-prepared hourly file -> UCI archive -> unchanged local hour.csv -> notebook tables and figures.

5. Basic data quality checks

Check	Result
Schema	17 expected columns; expected integer, float, and date-text types match.
Ranges	0 invalid values across the checked ranges.
Missingness	0 missing cells.
Exact duplicates	0 duplicate rows.
Date-hour duplicates	0 duplicate keys.
Date/month consistency	0 mismatches.
Anomalies	Minimum/maximum values, histogram, boxplot, and the five largest counts inspected.
Leakage	cnt equals casual + registered on every row; 0 failures.

Ranges checked: hr 0-23; mnth 1-12; season and weathersit 1-4; workingday and holiday 0/1; temp, atemp, hum, and windspeed 0-1; cnt nonnegative. No rows were removed or values filled.

5.1 Anomaly review and leakage

Capital Bikeshare, 2011-2012 (17,379 recorded hours)

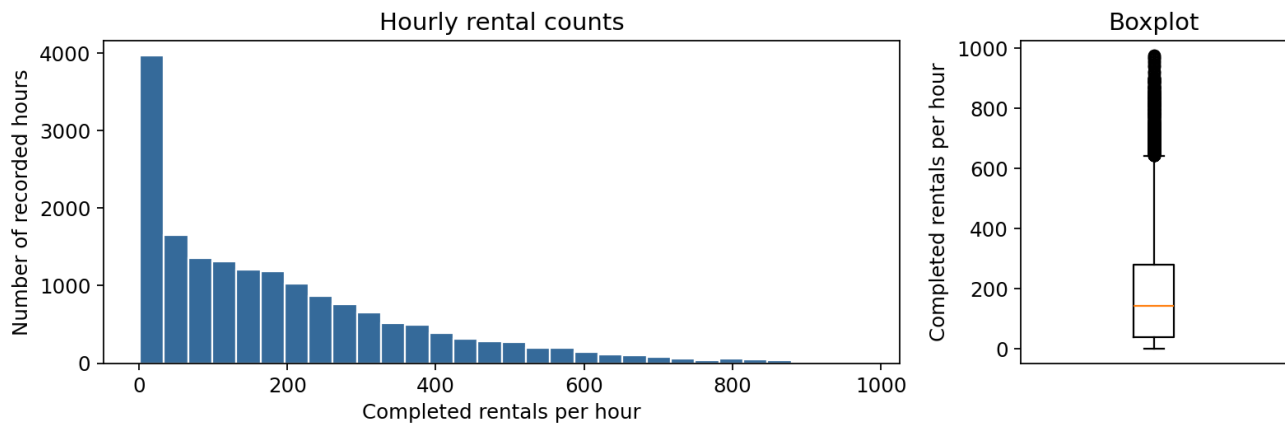


Figure 1. Rental-count histogram and boxplot for the 17,379 recorded hours. Source: the unchanged UCI hourly CSV.

Counts range from 1 to 977 rentals per hour. The distribution has a long right tail. Large counts may represent real peak periods, so they should not automatically be removed as errors. Inspecting the five largest counts does not prove that every record is error-free.

Simple summary statistics

Variable	Count	Mean	Median	Std. dev.
cnt	17,379	189.46	142.00	181.39
casual	17,379	35.68	17.00	49.31
registered	17,379	153.79	115.00	151.36
temp	17,379	0.497	0.500	0.193
atemp	17,379	0.476	0.485	0.172
hum	17,379	0.627	0.630	0.193
windspeed	17,379	0.190	0.194	0.122

Temperature, feeling temperature, humidity, and wind speed use the supplied normalized values. The standard deviation describes variation across recorded hours.

Leakage finding

The notebook directly checks **cnt == casual + registered**. The equality holds on every row. If cnt is the prediction target, casual and registered should not be used as predictors because together they directly reconstruct the target. No prediction model is trained here.

6. Rental patterns by hour and working day

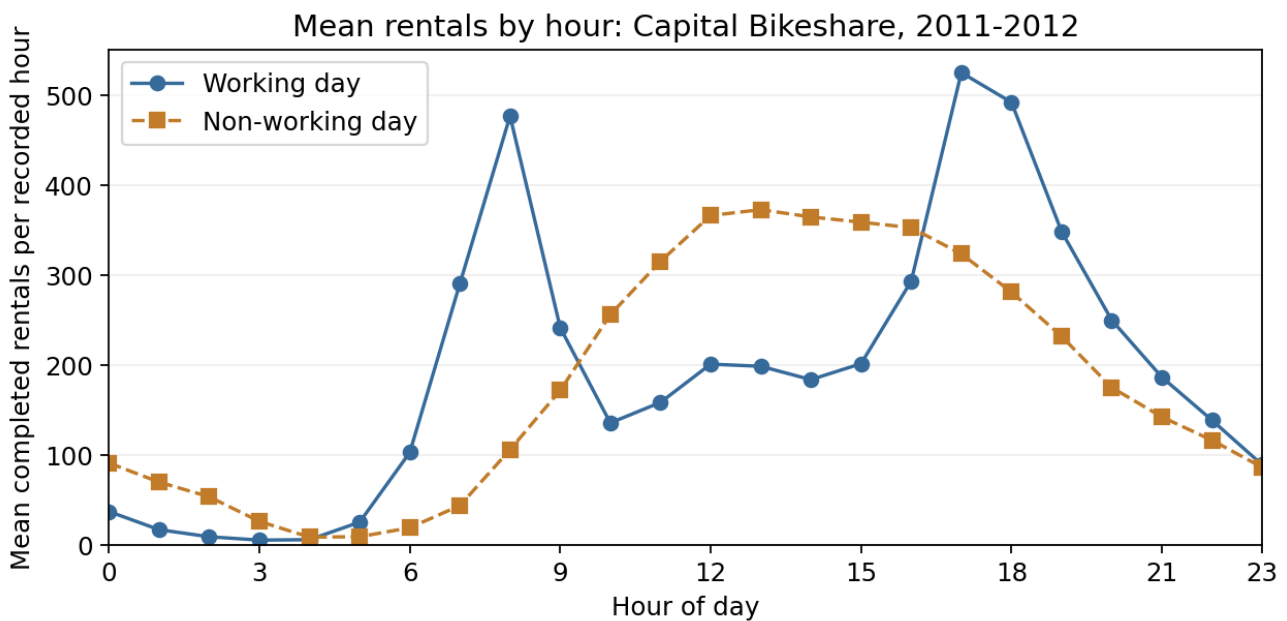


Figure 2. Mean completed rentals by hour, separately for working and non-working days. Each average uses recorded hours from 2011-2012.

Working days show morning and late-afternoon peaks. Their highest mean is at 17:00 (525.29 rentals). Non-working days have a broader daytime peak, with the highest mean at 13:00 (372.73 rentals). These patterns do not establish why people rented bikes.

Overall averages by working-day status

Day type	Recorded hours	Mean	Median	Std. dev.
Non-working day	5,514	181.41	119.00	172.85
Working day	11,865	193.21	151.00	185.11

The README defines `workingday = 1` when a day is neither a weekend nor a holiday. The overall means describe averages per recorded hour. Figure 2 shows that the timing of rentals also differs between the two groups.

7. Rental patterns by weather

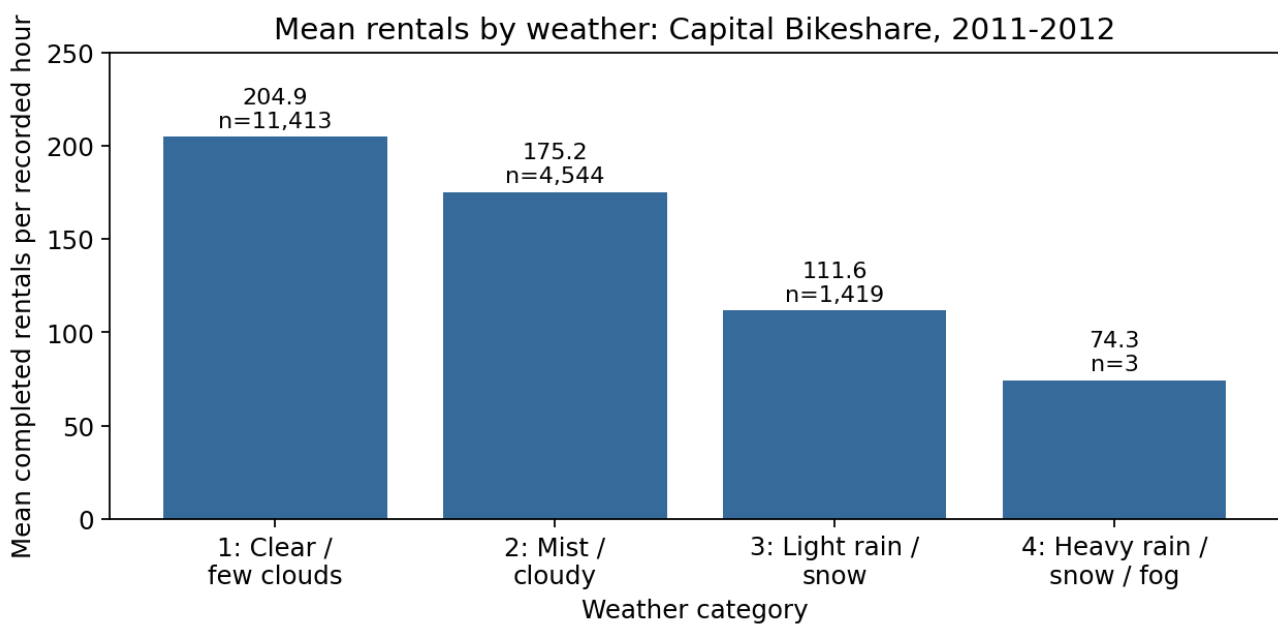


Figure 3. Mean completed rentals per recorded hour by weather category. n is the number of recorded hours in each category. Labels shorten the official README descriptions.

Weather code	Recorded hours	Mean	Median	Std. dev.
1	11,413	204.87	159.00	189.49
2	4,544	175.17	133.00	165.43
3	1,419	111.58	63.00	133.78
4	3	74.33	36.00	77.93

The mean is 204.87 rentals per hour in clear/few-cloud weather and 111.58 in light rain/snow. Category 4 has only 3 records, so its average should be read cautiously.

These averages describe associations. Weather groups may contain different seasons, hours, and working-day patterns, so the comparison does not establish a weather effect.

8. Bias and measurement limitations

1. The data come only from Capital Bikeshare. Other systems may have different patterns.
2. The data cover only 2011-2012. Current patterns may differ.
3. Weather and rentals have an observational association. Season, time of day, and working-day status may also affect the patterns.

Measurement counterexample

Suppose 200 people want to rent a bike, but only 100 bikes are available. In a simple example where each bike is rented once, about 100 completed rentals would be recorded rather than all 200 people who wanted a bike. Therefore, cnt measures completed rentals, not perfect latent demand.

This is a conceptual example, not an observed event in the dataset.

9. Prohibited claims

We should not claim that weather causes changes in bike rental demand based only on this observational dataset.

We should not assume that the results apply to all cities, all bike-sharing systems, or current users.

10. Conclusion

The basic checks found no missing values, duplicates, invalid values in the checked ranges, or count-identity failures. Rental patterns differ by hour, working-day status, and weather category. The averages describe this dataset and should not be interpreted as causal effects. Completed rentals do not measure all demand.

AI use and companion files

OpenAI Codex / ChatGPT helped organize the work, write and check code, and improve explanations and figures. The notebook was rerun in a fresh kernel and its main values were checked against the raw CSV. This does not claim personal manual verification by the student. The repository includes the executable notebook, Data Card, data dictionary, AI-use log, two result tables, and submission summary.

Sources

Fanaee-T, H. (2013). Bike Sharing [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C5W894. CC BY 4.0. Official source: UCI Bike Sharing page. The download URL and hourly SHA-256 are recorded in docs/data_card.md. Official README.txt, included with the dataset.

README-requested attribution: Fanaee-T and Gama (2013), Event labeling combining ensemble detectors and background knowledge. DOI: 10.1007/s13748-013-0040-3.

Repository: github.com/Zhoujie-SONG/DS-A1-bike-sharing-audit